

Namespace CodiceFiscale

Classes

[CodiceFiscaleGenerator](#)

Provides methods to generate an Italian Codice Fiscale.

[CodiceFiscaleMatcher](#)

Provides methods to match and compare Italian Codice Fiscale strings.

[CodiceFiscaleParser](#)

Provides methods to parse an Italian Codice Fiscale into its constituent data.

[CodiceFiscaleValidator](#)

Provides methods to validate Italian Codice Fiscale codes.

[IBANValidator](#)

Provides methods to validate IBAN (International Bank Account Number) strings. Only SEPA countries are supported (EU27 + UK, Switzerland, Norway, Iceland, Liechtenstein, Monaco, San Marino, Andorra, Vatican City).

[ItalianVatCodeValidator](#)

Provides validation for Italian VAT codes (Partita IVA) and fiscal codes used in a VAT context.

Class CodiceFiscaleGenerator

Namespace: [CodiceFiscale](#)

Assembly: CodiceFiscale.dll

Provides methods to generate an Italian Codice Fiscale.

```
public class CodiceFiscaleGenerator
```

Inheritance

[object](#)  ← CodiceFiscaleGenerator

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Methods

Generate(string, string, DateOnly, Gender)

Generates a Codice Fiscale based on the provided personal information.

```
public static string Generate(string name, string surname, DateOnly dateOfBirth,  
    Gender gender)
```

Parameters

name [string](#) 

The person's first name.

surname [string](#) 

The person's surname.

dateOfBirth [DateOnly](#) 

The person's date of birth.

gender [Gender](#)

The person's gender ('M' or 'F').

Returns

[string](#) 

The generated Codice Fiscale string.

Class CodiceFiscaleMatcher

Namespace: [CodiceFiscale](#)

Assembly: CodiceFiscale.dll

Provides methods to match and compare Italian Codice Fiscale strings.

```
public class CodiceFiscaleMatcher
```

Inheritance

[object](#)  ← CodiceFiscaleMatcher

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Class CodiceFiscaleParser

Namespace: [CodiceFiscale](#)

Assembly: CodiceFiscale.dll

Provides methods to parse an Italian Codice Fiscale into its constituent data.

```
public class CodiceFiscaleParser
```

Inheritance

[object](#)  ← CodiceFiscaleParser

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

CodiceFiscaleParser()

Provides methods to parse an Italian Codice Fiscale into its constituent data.

```
public CodiceFiscaleParser()
```

Methods

Parse(string)

Parses the specified Codice Fiscale and returns the decoded personal data.

```
public static CodiceFiscaleData Parse(string cf)
```

Parameters

cf [string](#) 

The Codice Fiscale string to parse. Case-insensitive.

Returns

[CodiceFiscaleData](#)

A [CodiceFiscaleData](#) instance containing the gender, date of birth, and Belfiore code extracted from the Codice Fiscale.

Exceptions

[InvalidCodiceFiscaleException](#)

Thrown when `cf` is not a valid Codice Fiscale.

TryParse(string, out CodiceFiscaleData?)

Attempts to parse the specified Codice Fiscale without throwing an exception.

```
public static bool TryParse(string cf, out CodiceFiscaleData? data)
```

Parameters

`cf` [string](#)

The Codice Fiscale string to parse. Case-insensitive.

`data` [CodiceFiscaleData](#)

When this method returns [true](#), contains a [CodiceFiscaleData](#) instance with the gender, date of birth, and Belfiore code extracted from the Codice Fiscale; otherwise, [null](#).

Returns

[bool](#)

[true](#) if `cf` was parsed successfully; otherwise, [false](#).

Examples

```
if (CodiceFiscaleParser.TryParse("RSSMRA85M01H501Z", out CodiceFiscaleData? data))
{
    Console.WriteLine($"Gender: {data.Gender}, Born: {data.DateOfBirth}");
}
```

Class CodiceFiscaleValidator

Namespace: [CodiceFiscale](#)

Assembly: CodiceFiscale.dll

Provides methods to validate Italian Codice Fiscale codes.

```
public class CodiceFiscaleValidator
```

Inheritance

[object](#)  ← CodiceFiscaleValidator

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Methods

IsValid(string)

Determines whether the specified Codice Fiscale is valid.

```
public static bool IsValid(string cf)
```

Parameters

cf [string](#) 

The Codice Fiscale string to validate.

Returns

[bool](#) 

true if the Codice Fiscale is valid; otherwise, **false**.

Class IBANValidator

Namespace: [CodiceFiscale](#)

Assembly: CodiceFiscale.dll

Provides methods to validate IBAN (International Bank Account Number) strings. Only SEPA countries are supported (EU27 + UK, Switzerland, Norway, Iceland, Liechtenstein, Monaco, San Marino, Andorra, Vatican City).

```
public class IBANValidator
```

Inheritance

[object](#)  ← IBANValidator

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Methods

IsValid(string)

Determines whether the specified IBAN string is valid. Spaces are ignored. Only SEPA country codes are accepted.

```
public static bool IsValid(string iban)
```

Parameters

iban [string](#) 

The IBAN string to validate. Spaces are allowed and will be stripped.

Returns

[bool](#) 

true if the IBAN is valid and belongs to a SEPA country; otherwise, **false**.

Class ItalianVatCodeValidator

Namespace: [CodiceFiscale](#)

Assembly: CodiceFiscale.dll

Provides validation for Italian VAT codes (Partita IVA) and fiscal codes used in a VAT context.

```
public class ItalianVatCodeValidator
```

Inheritance

[object](#)  ← ItalianVatCodeValidator

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Methods

IsValid(string, bool, bool)

Validates the specified Italian VAT or fiscal code string.

```
public static bool IsValid(string vat, bool isConsumer, bool isFiscal)
```

Parameters

vat [string](#) 

The code to validate. Can be an 11-digit Partita IVA or, if **isFiscal** is **true**, a 16-character Codice Fiscale.

isConsumer [bool](#) 

true if the code belongs to a natural person (consumer); the first digit must be 8 or 9. **false** for standard company VAT codes (first digit 0–7).

isFiscal [bool](#) 

`true` to also accept a 16-character Codice Fiscale as valid input in addition to an 11-digit VAT code.
`false` to accept only the 11-digit Partita IVA format.

Returns

`bool`[↗](#)

`true` if the code is valid according to the specified rules; otherwise, `false`.

Namespace CodiceFiscale.Entities

Classes

[CodiceFiscaleData](#)

Represents the data required to generate or parse an Italian Codice Fiscale.

[Coordinates](#)

Represents a geographical coordinate with latitude and longitude.

[ForeignCountry](#)

Represents a foreign country as encoded in the Italian Codice Fiscale dataset.

[Municipality](#)

Represents an Italian municipality with related administrative and contact information.

[Province](#)

Represents an Italian province with its name, acronym, code, and region.

Class CodiceFiscaleData

Namespace: [CodiceFiscale.Entities](#)

Assembly: CodiceFiscale.dll

Represents the data required to generate or parse an Italian Codice Fiscale.

```
public record CodiceFiscaleData : IEquatable<CodiceFiscaleData>
```

Inheritance

[object](#)  ← CodiceFiscaleData

Implements

[IEquatable](#)  <[CodiceFiscaleData](#)>

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

CodiceFiscaleData(Gender, DateOnly, string)

Represents the data required to generate or parse an Italian Codice Fiscale.

```
public CodiceFiscaleData(Gender Gender, DateOnly DateOfBirth, string BelfioreCode)
```

Parameters

Gender [Gender](#)

The gender of the individual ("Male" for male, "Female" for female).

DateOfBirth [DateOnly](#) 

The date of birth of the individual.

BelfioreCode [string](#) 

The Belfiore code representing the place of birth.

Properties

BelfioreCode

The Belfiore code representing the place of birth.

```
public string BelfioreCode { get; init; }
```

Property Value

[string](#)

DateOfBirth

The date of birth of the individual.

```
public DateOnly DateOfBirth { get; init; }
```

Property Value

[DateOnly](#)

Gender

The gender of the individual ("Male" for male, "Female" for female).

```
public Gender Gender { get; init; }
```

Property Value

[Gender](#)

Class Coordinates

Namespace: [CodiceFiscale.Entities](#)

Assembly: CodiceFiscale.dll

Represents a geographical coordinate with latitude and longitude.

```
public record Coordinates : IEquatable<Coordinates>
```

Inheritance

[object](#)  ← Coordinates

Implements

[IEquatable](#)  <[Coordinates](#)>

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

Coordinates(double, double)

Represents a geographical coordinate with latitude and longitude.

```
public Coordinates(double Lat, double Lng)
```

Parameters

Lat [double](#) 

Gets the latitude component of the coordinate.

Lng [double](#) 

Gets the longitude component of the coordinate.

Properties

Lat

Gets the latitude component of the coordinate.

```
[JsonProperty("lat")]  
public double Lat { get; init; }
```

Property Value

[double](#)

Lng

Gets the longitude component of the coordinate.

```
[JsonProperty("lng")]  
public double Lng { get; init; }
```

Property Value

[double](#)

Class ForeignCountry

Namespace: [CodiceFiscale.Entities](#)

Assembly: CodiceFiscale.dll

Represents a foreign country as encoded in the Italian Codice Fiscale dataset.

```
public record ForeignCountry : IEquatable<ForeignCountry>
```

Inheritance

[object](#)  ← ForeignCountry

Implements

[IEquatable](#)  <[ForeignCountry](#)>

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

ForeignCountry(string, string, string, bool)

Represents a foreign country as encoded in the Italian Codice Fiscale dataset.

```
public ForeignCountry(string CatastalCode, string Name, string Continent, bool Deleted)
```

Parameters

CatastalCode [string](#) 

The Belfiore catastal code assigned to the country (e.g. **Z614** for Venezuela). All foreign country codes start with the letter **Z**.

Name [string](#) 

The Italian name of the country (e.g. **Venezuela**).

Continent [string](#) 

The continent to which the country belongs (e.g. **America**).

Deleted [bool](#)

true if the country code has been abolished (e.g. former states such as Czechoslovakia); **false** for currently active countries.

Properties

CatastalCode

The Belfiore catastal code assigned to the country (e.g. **Z614** for Venezuela). All foreign country codes start with the letter **Z**.

```
[JsonProperty("codiceCatastale")]  
public string CatastalCode { get; init; }
```

Property Value

[string](#)

Continent

The continent to which the country belongs (e.g. **America**).

```
[JsonProperty("continente")]  
public string Continent { get; init; }
```

Property Value

[string](#)

Deleted

true if the country code has been abolished (e.g. former states such as Czechoslovakia); **false** for currently active countries.

```
[JsonProperty("soppresso")]  
public bool Deleted { get; init; }
```

Property Value

[bool](#)

Name

The Italian name of the country (e.g. *Venezuela*).

```
[JsonProperty("nome")]  
public string Name { get; init; }
```

Property Value

[string](#)

Class Municipality

Namespace: [CodiceFiscale.Entities](#)

Assembly: CodiceFiscale.dll

Represents an Italian municipality with related administrative and contact information.

```
public record Municipality : IEquatable<Municipality>
```

Inheritance

[object](#)  ← Municipality

Implements

[IEquatable](#)  <[Municipality](#)>

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

CatastalCode

Gets the catastal code of the municipality.

```
[JsonProperty("codiceCatastale")]  
public required string CatastalCode { get; init; }
```

Property Value

[string](#) 

Code

Gets the unique code of the municipality.

```
[JsonProperty("codice")]  
public string? Code { get; init; }
```

Property Value

[string](#) 

Coordinates

Gets the geographical coordinates of the municipality.

```
[JsonProperty("coordinate")]  
public Coordinates? Coordinates { get; init; }
```

Property Value

[Coordinates](#)

Email

Gets the email address of the municipality.

```
[JsonProperty("email")]  
public string? Email { get; init; }
```

Property Value

[string](#) 

FAX

Gets the fax number of the municipality.

```
[JsonProperty("fax")]  
public string? FAX { get; init; }
```

Property Value

[string](#)

ForeignName

Gets the foreign name of the municipality, if available.

```
[JsonProperty("nomeStraniero")]  
public string? ForeignName { get; init; }
```

Property Value

[string](#)

Name

Gets the name of the municipality.

```
[JsonProperty("nome")]  
public string? Name { get; init; }
```

Property Value

[string](#)

PEC

Gets the certified email (PEC) address of the municipality.

```
[JsonProperty("pec")]  
public string? PEC { get; init; }
```

Property Value

[string](#)

Prefix

Gets the telephone prefix of the municipality.

```
[JsonProperty("prefisso")]  
public string? Prefix { get; init; }
```

Property Value

[string](#)[↗]

Province

Gets the province to which the municipality belongs.

```
[JsonProperty("provincia")]  
public Province? Province { get; init; }
```

Property Value

[Province](#)

Telephone

Gets the telephone number of the municipality.

```
[JsonProperty("telefono")]  
public string? Telephone { get; init; }
```

Property Value

[string](#)[↗]

ZipCode

Gets the ZIP code of the municipality.

```
[JsonProperty("cap")]  
public string? ZipCode { get; init; }
```

Property Value

[string](#) 

Class Province

Namespace: [CodiceFiscale.Entities](#)

Assembly: CodiceFiscale.dll

Represents an Italian province with its name, acronym, code, and region.

```
public record Province : IEquatable<Province>
```

Inheritance

[object](#)  ← Province

Implements

[IEquatable](#)  <[Province](#)>

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

Province(string, string, string, string)

Represents an Italian province with its name, acronym, code, and region.

```
public Province(string Name, string Acronym, string Code, string Region)
```

Parameters

Name [string](#) 

The name of the province.

Acronym [string](#) 

The acronym (sigla) of the province.

Code [string](#) 

The code of the province.

Region [string](#)

The region to which the province belongs.

Properties

Acronym

The acronym (sigla) of the province.

```
[JsonProperty("sigla")]  
public string Acronym { get; init; }
```

Property Value

[string](#)

Code

The code of the province.

```
[JsonProperty("codice")]  
public string Code { get; init; }
```

Property Value

[string](#)

Name

The name of the province.

```
[JsonProperty("nome")]  
public string Name { get; init; }
```

Property Value

[string](#) 

Region

The region to which the province belongs.

```
[JsonProperty("regione")]  
public string Region { get; init; }
```

Property Value

[string](#) 

Namespace CodiceFiscale.Enums

Enums

[Gender](#)

Specifies the biological gender of an individual, as encoded in the Italian Codice Fiscale.

Enum Gender

Namespace: [CodiceFiscale.Enums](#)

Assembly: CodiceFiscale.dll

Specifies the biological gender of an individual, as encoded in the Italian Codice Fiscale.

```
public enum Gender : byte
```

Fields

Female = 2

Represents a female individual. The day-of-birth field in the Codice Fiscale ranges from 41 to 71 (actual day + 40).

Male = 1

Represents a male individual. The day-of-birth field in the Codice Fiscale ranges from 1 to 31.

Namespace CodiceFiscale.Exceptions

Classes

[InvalidCodiceFiscaleException](#)

The exception that is thrown when an Italian Codice Fiscale string fails validation.

Class InvalidCodiceFiscaleException

Namespace: [CodiceFiscale.Exceptions](#)

Assembly: CodiceFiscale.dll

The exception that is thrown when an Italian Codice Fiscale string fails validation.

[Serializable]

```
public class InvalidCodiceFiscaleException : Exception, ISerializable
```

Inheritance

[object](#) ← [Exception](#) ← InvalidCodiceFiscaleException

Implements

[ISerializable](#)

Inherited Members

[Exception.GetBaseException\(\)](#), [Exception.GetType\(\)](#), [Exception.ToString\(\)](#), [Exception.Data](#), [Exception.HelpLink](#), [Exception.HResult](#), [Exception.InnerException](#), [Exception.Message](#), [Exception.Source](#), [Exception.StackTrace](#), [Exception.TargetSite](#), [Exception.SerializeObjectState](#), [object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#)

Constructors

InvalidCodiceFiscaleException()

Initializes a new instance of [InvalidCodiceFiscaleException](#) with no message.

```
public InvalidCodiceFiscaleException()
```

InvalidCodiceFiscaleException(string)

Initializes a new instance of [InvalidCodiceFiscaleException](#) with a descriptive message.

```
public InvalidCodiceFiscaleException(string message)
```

Parameters

message [string](#)

A human-readable message that describes the validation failure.

InvalidCodiceFiscaleException(string, Exception)

Initializes a new instance of [InvalidCodiceFiscaleException](#) with a descriptive message and a reference to the inner exception that caused this exception.

```
public InvalidCodiceFiscaleException(string message, Exception innerException)
```

Parameters

message [string](#)

A human-readable message that describes the validation failure.

innerException [Exception](#)

The exception that is the cause of the current exception.